

Ryan Li

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EDUCATION

University of Waterloo

Waterloo, ON

Computer Science and Combinatorics & Optimization (CO) Double Major

- WAT.ai, Computer Science Club, Data Science Club, Quant Club **GPA: 96.2% (97.1% Faculty Average)**

TECHNICAL SKILLS

Languages: Python, C/C++, Java, SQL, JavaScript, TypeScript, Go, Rust

ML/Data: PyTorch, scikit-learn, Hugging Face, NumPy, SciPy, pandas, OpenCV, Pillow, Airflow, W&B, Modal

Web/Frameworks: Flask, FastAPI, Node.js/Express.js, React

Infra/Tools: Git, Docker, Kubernetes, Redis, Prometheus, Postgres, MariaDB, MongoDB, AWS, GCP, GitHub Actions

HONORS & AWARDS

- **National Mathematics Scholarship (\$15,000)**, President's Scholarship of Distinction (\$2,000)
- **Governor General's Medal** (graduated 1/400+), IB Diploma 44/45 (**top ~0.5% worldwide**)
- **Canadian Mathematics Olympiad Qualifier, Fermat Winner (placed 1/19644)**, 3× AIME, Putnam Top 25%

WORK EXPERIENCE

Software Engineer Intern

May 2026 – Aug 2026

Environment and Climate Change Canada (Government of Canada)

- **Spearheaded** the Work Planning Tool used by **300+ field technicians** across Canada; engineered **ETL pipelines** to sync Aquarius hydrometric data into a Flask/MariaDB backend with AWS S3-backed storage.
- Designed a DAG longest-path algorithm (topological sort + dynamic programming) to reconcile overlapping Aquarius approval entries, maintaining consistent records over **2,800+ monitoring stations**.
- Architected an **agentic harness** (LangGraph, AWS Bedrock) that dynamically routes between query-planning and tool-call nodes via conditional edges to automate data queries for technicians.
- Deployed Prometheus monitoring and introduced Docker layer caching to CI/CD, **cutting release time by 85%**.

Machine Learning Researcher

May 2026 – Present

WAT.ai (University of Waterloo)

- Implemented a **video pseudo-labeling pipeline** for object state-change segmentation with shell-scripted batch runs across verbs; **closed 55% of the gap** between a no-pseudo-label baseline and the oracle ceiling (mIoU 0.503).
- Orchestrated multi-stage VLM inference (e.g. GroundingDINO → SAM → DeAOT → CLIP scoring) in **Apache Airflow**, tracking experiments with Weights & Biases for reproducible results across a 7-member team.

PROJECTS

AlphaHex - Catan Engine | Python, PyTorch, Docker, FastAPI, React, Vercel, Postgres, Supabase

- Trained a 1v1 Catan engine with a **Dockerized** self-play Monte Carlo Tree Search-NN pipeline (replay buffers, gated checkpoint acceptance, leaderboard evaluation), achieving a **top 100 Canada rank** on colonist.io.
- Deployed a React/FastAPI app for bot gameplay with OAuth sign-in and **Postgres (Supabase)** game history.

Slotify (UofTHacks Winner) | Python, TypeScript, Node.js, React, FFmpeg, Docker, Redis, BullMQ

- Built a React/Node app and API for inserting natural sponsor reads into podcasts using OpenAI ad generation and ElevenLabs voice cloning, backed by **Redis** for async BullMQ job queuing and caching audio/ad statements.
- **Reduced processing latency 35%** by offloading LLM use with FFmpeg/librosa/pydub pipelines for LUFS-loudness matching and beat-synced ad placement; placed **1/150+** teams at UofTHacks 13.

Provenance (cuHacking Winner) | TypeScript, Solana, MongoDB Atlas, Chrome Extension (MV3), Vercel

- Developed an **image authenticity system** that cryptographically signs photos and anchors them **on-chain** (Solana), achieving sub-second (~400ms) trustless verification in one RPC; placed **1/100+** teams at cuHacking 7.
- Extended verification beyond hash-matching to surfacing visually similar reposts via **MongoDB Vector Search**; set up **CI (GitHub Actions) unit testing** to ensure the client and on-chain program agree on data layout.